SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

QUESTIONS: 5 Total Marks:50

Annexure A

Scenario-Based

Code of Conduct:

I HARISHKUMAR SV (192511486) *(Name / Reg* *No)certify that this submission is my original work and*

that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary

action. ***Signature of the student:HARISH KUMAR SV***

Annexure A

Scenario-Based

1. A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already

been assigned:

School of Computing → /25

School of Electronics → /26

School of Mechanical Engineering → /27

Asthe network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department. 2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet. 4. Determine the remaining unused IP addressrange within the given network block.

2. A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departmentsrequire different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**

Quality Assurance → **/25**

Human Resources → **/26**

Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3.A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

2001:0db8:abcd:0000:0000:0000:0000:0000/64

As the network engineer responsible for the migration, perform the following: 1. Convert the given IPv6 address into its **compressed notation**. 2. Expand the compressed address back into its **full hexadecimal representation**.

3. Identify the **network prefix** using the **/64** prefix length.

4. Calculate the **total number of host addresses** supported within the subnet. 5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

192.168.1.0/24

192.168.2.0/24

192.168.3.0/24

A packet arrives at the router with the destination IP address:

192.168.2.45

As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.
3. The **next-hop network/interface** used to forward the packet.
4. The **default route** that should be used if the destination does not match any routing table entry.

5. A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN. 2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN. 4. Identify the **default gateway** that hosts in each LAN should use for communication.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understanding of the scenario and context	Shows good understanding with minor gaps	Partial understanding; misses key aspects	Misinterprets or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to addressthe scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts

Decision Making	20%	Makes well reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justificatio n	Makes weak decisions with limited justificatio n	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorgani ze d and difficult to understa nd

Course Outcome Covered

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology.

1. Question 1: Smart Campus Network Design

1.1 Scenario Explanation

Anewly established Smart Campus is deploying a secure network infrastructure forits academic and administrative departments. The network administrator has been allocated the IP address block . Based on the expected number of devices, specific

192.168.100.0/24

subnet masks have been assigned to three departments: School of Computing ( ), School of Electronics ( ), and School of Mechanical Engineering ( ). The task involves designing the subnet allocation, identifying network and broadcast addresses, valid host ranges, usable host counts, and determining any remaining unused IP address ranges within the allocated block.

1.2 Network Design Approach

The design approach for this scenario involves applying Variable Length Subnet Masking (VLSM) principles to efficiently allocate IP addresses. The subnets are allocated sequentially from the given block, prioritizing larger subnets first to minimize

192.168.100.0/24

fragmentation, although in this specific case, the orderis given by the problem. Each department receives a dedicated subnet, ensuring logical separation and efficient resource utilization. The calculations will determine the network address, broadcast address, valid hostrange, and the number ofusable hosts for each subnet. Finally, any unallocated IP space within the original block will be identified.

1.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:**
- **Number of Usable Hosts:** (subtracting network and broadcast addresses)

Working

Given Base Network: 192.168.100.0/24

1. School of Computing (SoC) - /25

- **Subnet Mask:** (255.255.255.128)
- **NumberofAddresses:**
- **Network Address** 192.168.100.0
- **Broadcast Address:** (192.168.100.0 + 128 - 1) 192.168.100.127
- **Valid Host Range:** 192.168.100.1 to 192.168.100.126
- **Usable Hosts:**

2. School of Electronics (SoE) - /26

This subnet will be allocated from the next available block after SoC. The next available address after 192.168.100.127 is

. 192.168.100.128

- **Subnet Mask:** (255.255.255.192)

- **Number of Addresses:**
- **Network Address**[redacted] 192.168.100.128

- **Broadcast Address:** 192.168.100.191 (192.168.100.128 + 64 - 1)

- **Valid Host Range:** • **Usable Hosts:** 192.168.100.190 to

192.168.100.129

3. School of Mechanical Engineering (SoME) - /27

This subnet will be allocated from the next available block after SoE. The next available address after is . 192.168.100.192

192.168.100.191

- **Subnet Mask:** (255.255.255.224)

- **Number of Addresses:**
- **Network Address**[redacted] 192.168.100.192

- **Broadcast Address:** 192.168.100.223 (192.168.100.192 + 32 - 1)

- **Valid Host Range:** • **Usable Hosts:** 192.168.100.222 to

Final Answer

192.168.100.193

Subnet Allocation Summary:

	Network Address	Subnet Mask	Broadcast Address		Usable Hos
School of Computing	192.168.100.0	/25	192.168.100.127	192.168.100.1 - 192.168.100.126	126
School of Electronics	192.168.100.128	/26	192.168.100.191	192.168.100.129 - 192.168.100.190	62
School of Mechanical Engineering	192.168.100.192	/27	192.168.100.223	192.168.100.193 - 192.168.100.222	30

Remaining Unused IP Address Range:

192.168.100.248/29 and 192.168.100.252/30 and 192.168.100.254/31 and 192.168.100.255/32 . More simply, it is 192.168.100.224 to 192.168.100.255 .

1.4 Network Diagram

Figure 1.1: Smart Campus Network Topology

1.5 IP Address Table

This table provides a detailed breakdown of the IP addressing scheme for each department.

Department	Network Address	First Host IP	La
School of Computing	192.168.100.0	192.168.100.1	192
School of Electronics	192.168.100.128	192.168.100.129	192
School of Mechanical Engineering	192.168.100.192	192.168.100.193	192

1.6 Subnet Table

Subnet ID	Network Address	CIDR	Subnet Mask
1	192.168.100.0	/25	255.255.255.128
2	192.168.100.128	/26	255.255.255.192

3	192.168.100.1 92	/27	255. 126

1.7 Verification

62

CIDR

/25

30

/26

/27

Usable Hosts

The calculations were verified using the Python module, which confirms the ipaddress network, broadcast, and host ranges for each subnet. The sequential allocation ensures no overlap between subnets and efficient use of the IP address space. The remaining unused range was also confirmed by the script.

1.8 Conclusion

This section successfully designed the IP addressing scheme for the Smart Campus, allocating specific subnets to each department based on their requirements. The use of VLSM ensures optimal utilization of the block, with a clear understanding

192.168.100.0/24

of network boundaries and available host addresses. The identified unused range provides flexibility for future expansion.

2. Question 2: Multinational Software Company

Network Expansion

2.1 Scenario Explanation

Amultinational software company is expanding its headquarters and requires a newIP addressing scheme forits departments using the private IP block [redacted] 10.10.0.0/16 . Different departments have varying subnet size requirements: Software Development([redacted] /24), Quality Assurance ( ), Human Resources ( ), and Executive Management (/27 [redacted]. The task is to determine the network and broadcast addresses, valid host ranges, usable host counts for each department, and identify any remaining unused IP address ranges within the allocated block.

2.2 Network Design Approach

Similar to Question 1, this scenario employs VLSM to segment the 10.10.0.0/16 private IP block. The allocation prioritizes the largest subnet ( ) first, followed by /25 , , and to ensure efficient use ofthe address space and to avoid fragmentation. Each department receives a dedicated subnet, facilitating network management, security, and performance. The calculations will provide all necessary IP addressing details and identify the unused portions of the block.

10.10.0.0/16

2.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:**
- **Number of Usable Hosts:**

Working

10.10.0.0/16

Given Base Network:

1. Software Development (SWD) - /24 •

Subnet Mask: (255.255.255.0) •

NumberofAddresses:

- **Network Address** [redacted]

- **Broadcast Address:**

10.10.0.255

- **Valid Host Range:** to
-

Usable Hosts:

2. Quality Assurance (QA) - /25

10.10.0.254

This subnet will be allocated from the next available block, which is . We

10.10.1.0/24

then subnet this block.

- **Subnet Mask:** (255.255.255.128)
- **NumberofAddresses:**
- **Network Address** [redacted]
- **Broadcast Address:**
- **Valid Host Range:** to
- **Usable Hosts:**

3. Human Resources (HR) - /26

This subnet will be allocated from the remaining portion of after QA. The next

10.10.1.0/24

available address after is .

- **Subnet Mask:** (255.255.255.192)
- **NumberofAddresses:**
- **Network Address** [redacted]
- **Broadcast Address:**
- **Valid Host Range:** to
- **Usable Hosts:**

4. Executive Management (Exec) - /27

This subnet will be allocated from the remaining portion of after HR. The next

10.10.1.0/24

available address after is .

- **Subnet Mask:**

(255.255.255.224) •

NumberofAddresses:

- **Network** **Address**[redacted]
- **Broadcast Address:**
- **Valid Host** **Subnet Allocation**

Range: to • **Usable** **Summary:**

Hosts: 10.10.1.222

Final Answer

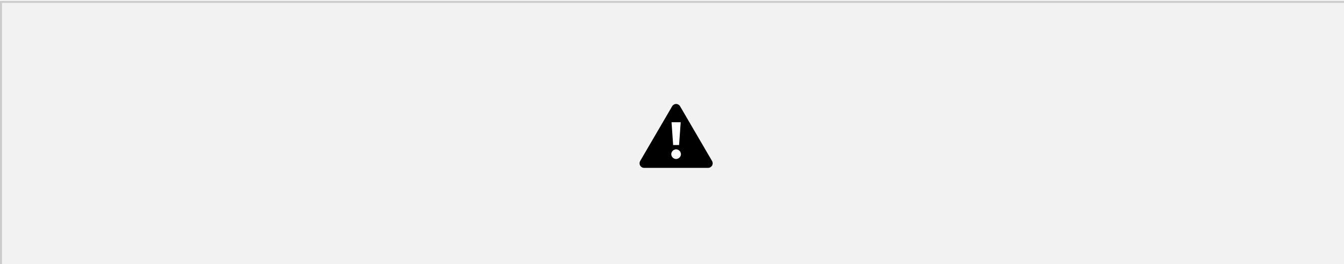
Department	Network Address	Subnet Mask	Usable Hos
Software Development	10.10.0.0	/24	254
Quality Assurance	10.10.1.0	/25	126
Human Resources	10.10.1.128	/26	62
Executive Management	10.10.1.192	/27	30

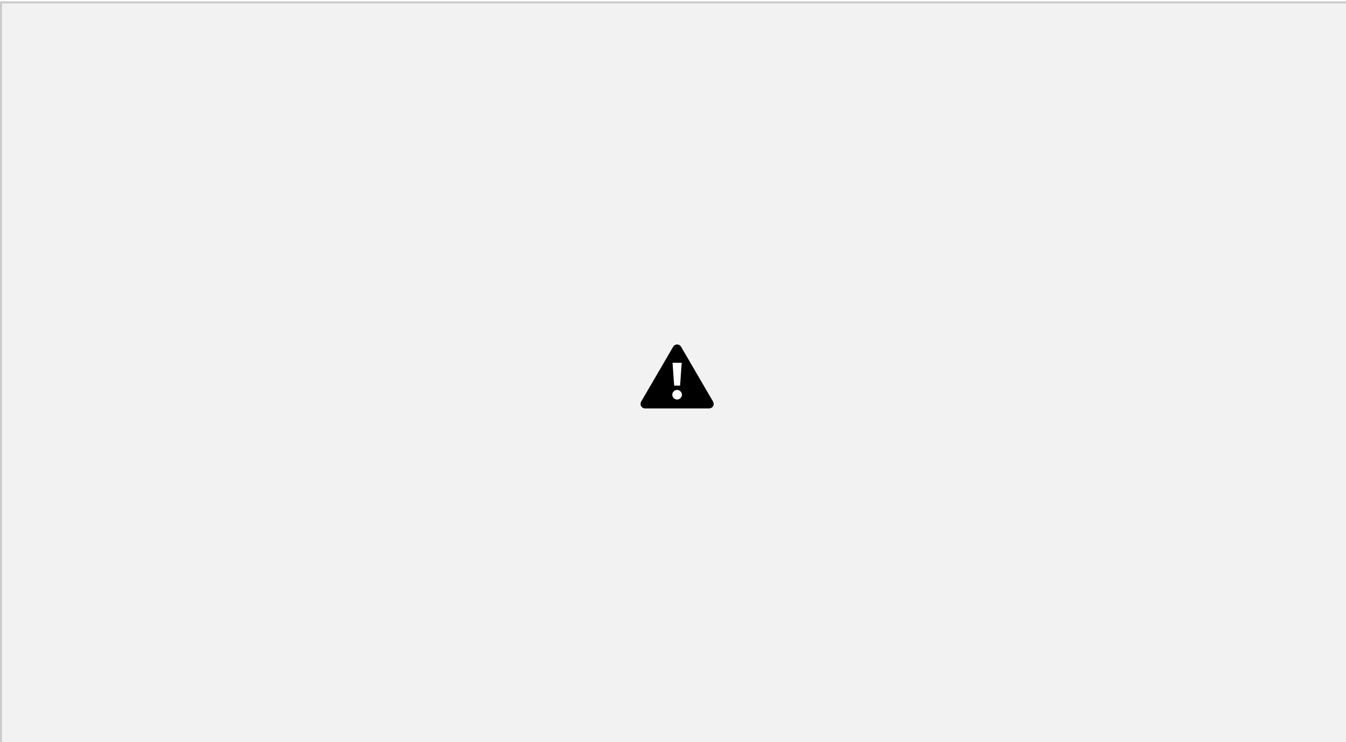
Remaining Unused IP Address Range:

- Within the 10.10.1.0/24 block, the range to remains unused. This 10.10.1.224 corresponds to 10.10.1.224/27 and subsequent smaller blocks.
- The entire range from also unused. to within the original block is 10.10.255.255 10.10.0.0/16

2.4 Network Diagram

Figure 2.1: Multinational Software Company Corporate Network Topology





2.5 IP Address Table

Department	Network Address	First Host IP	La
Software Developm en t	10.10.0.0	10.10.0.1	
Quality Assurance	10.10.1.0	10.10.1.1	
Human Resources	10.10.1.128	10.10.1.129	
Executive Manage men t	10.10.1.192	10.10.1.193	10.10.1.222

2.6 Subnet Table

CIDR

/24

/25 /27

/26

Network Address

Usable Hosts

Subnet ID

1	10.10.0.0	/24	255.255.255.0	10.10.0.255	10.10.0.1 - 10.10.0.254	254
2	10.10.1.0	/25	255.255.255.128	10.10.1.127	10.10.1.1 - 10.10.1.126	126
3	10.10.1.128	/26	255.255.255.192	10.10.1.191	10.10.1.129 - 10.10.1.190	62
4	10.10.1.192	/27	255.255.255.224	10.10.1.223	10.10.1.193 - 10.10.1.222	30

2.7 Verification

The calculations were verified using the ipaddress Python module, confirming the network, broadcast, and host ranges for each subnet. The sequential allocation within the

10.10.1.0/24 unused and subsequent 10.10.2.0/23 block and the identification of the larger blocks within the space were also confirmed. 10.10.0.0/16

2.8 Conclusion

This section successfully designed the IP addressing scheme for the multinational software company's headquarters, utilizing VLSM to efficiently allocate subnets to various departments. The detailed breakdown ofnetwork parameters and the identification of unused IP ranges provide a robust and scalable foundation forthe company's network infrastructure.

3. Question 3: IPv6 Addressing for Smart Hospital

3.1 Scenario Explanation

Asmart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network . The task involves converting the 2001:0db8:abcd:0000:0000:0000:0000:0000/64 given IPv6 address to compressed notation, expanding it back to full hexadecimal representation, identifying the network prefix, and calculating the total number ofhost addresses supported within the subnet.

3.2 Network Design Approach

This section focuses on understanding and manipulating IPv6 addresses, specifically compression, expansion, and host address calculation. IPv6 uses a 128-bit address space,
significantly larger than IPv4, to accommodate the vast number of devices in modern networks like a smart hospital. The prefix length is standard for most IPv6 subnets, dividing the address into a 64-bit network prefix and a 64-bit interface identifier. This approach ensures proper addressing for a large number ofdevices.

3.3 Step-by-step Calculations

Formula Used

- **Total Host Addresses:**

Working

Given IPv6 Network: 2001:0db8:abcd:0000:0000:0000:0000:0000/64

1. Convert to Compressed Notation:

- Rule: Consecutive blocks of zeros can be compressed with per address. . Only one is allowed

- Original: 2001:0db8:abcd:0000:0000:0000:0000:0000

- Compressed

2. Expand Compressed Address:

- Rule: Replace with enough blocks to make the address 8 hextets long. 0000
- Compressed:

- Expanded: 2001:0db8:abcd:0000:0000:0000:0000:0000

3. Identify Network Prefix:

- Given prefix length:
- The first 64 bits (or 4 hextets) constitute the network prefix.
- Network Prefix: or

2001:0db8:abcd:0000/64 2001:db8:abcd:0::/64

4. Calculate Total Number of Host Addresses:

- Total bits in IPv6 address: 128
- Host bits: 64
- Prefix length:

Answer

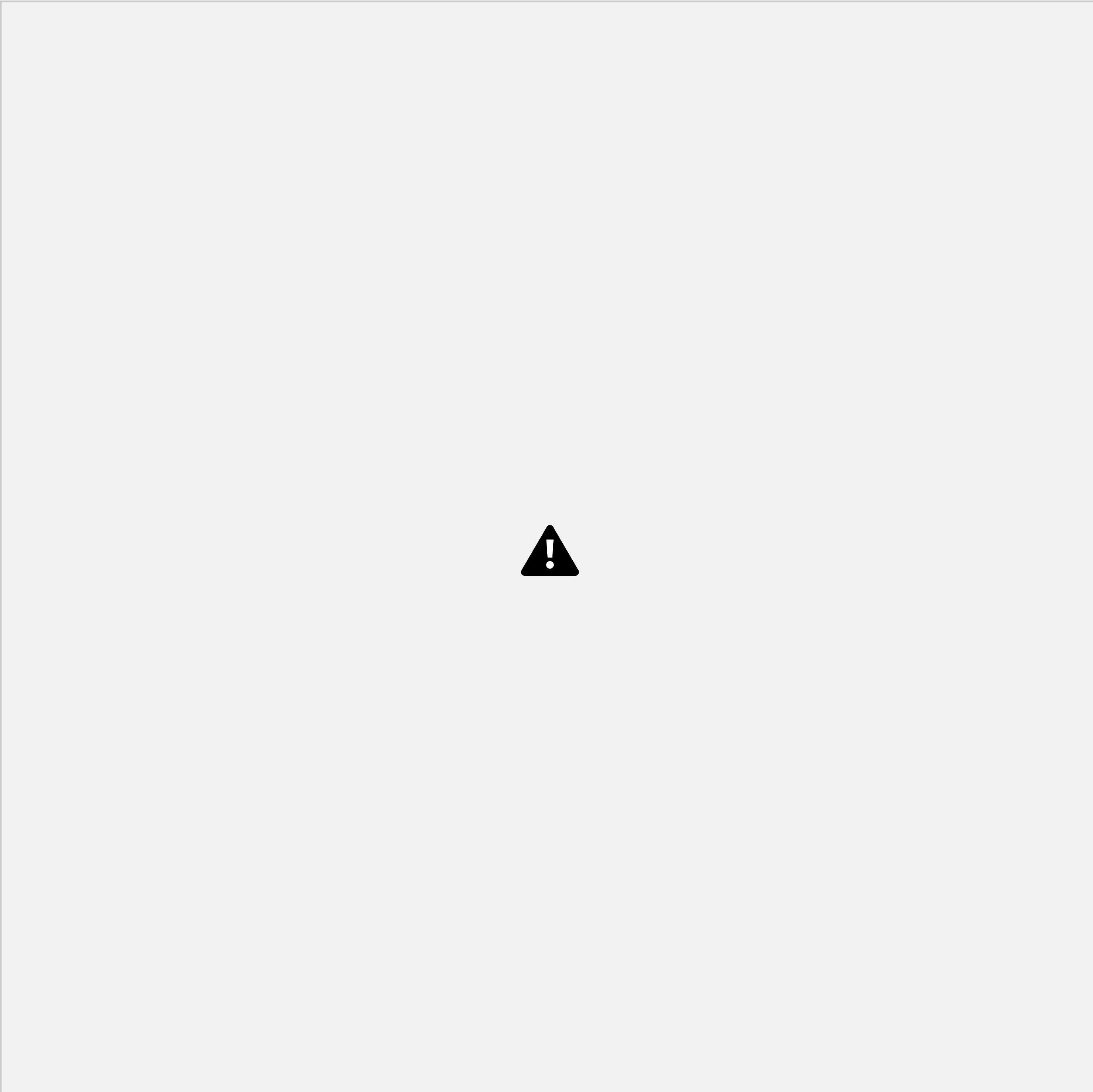
- Total Host Addresses: **Final**

2^64 = 18,446,744,073,709,551,616

- **Compressed Notation:** 2001:db8:abcd::/64
- **Expanded Notation:** 2001:0db8:abcd:0000:0000:0000:0000:0000/64
- **NetworkPrefix:** 2001:db8:abcd:0::/64
- **Total Number of Host Addresses:** 18,446,744,073,709,551,616

3.4 IPv6 Addressing Diagram

Figure 3.1: IPv6 Address Structure and Compression



3.5 Verification

The IPv6 compression and expansion were verified against standard IPv6 notation rules. The calculation for the total number of host addresses is a direct application of the formula , which is for a prefix. The Python script also confirmed these values.

bits) 2^64

3.6 Conclusion

This section demonstrated a clear understanding of IPv6 addressing, including its compression and expansion rules, and the calculation of available host addresses within a given subnet. The vast address space of IPv6 is crucial for supporting the immense number ofdevices in modern IoT-driven environments like a smart hospital.

4. Question 4: Router Packet Forwarding

4.1 Scenario Explanation

An Internet Service Provider (ISP) router has three routing table entries: ,

192.168.1.0/24

192.168.2.0/24 , and 192.168.3.0/24 . Apacket arrives at the router with the destination IP address 192.168.2.45 . The task is to determine the destination subnet, the matching routing table entry, the next-hop network/interface for forwarding, and the default route ifno match is found.

4.2 Network Design Approach

This scenario illustrates the fundamental process ofpacket forwarding in a router. When a packet arrives, the router examines its destination IP address and compares it against entries in its routing table. The longest prefix match rule is applied to find the most specific route. Ifno specific route matches, the packet is forwarded via the defaultroute, if configured. This approach ensures that packets are directed to their correct destination or, failing that, to a gateway that can furtherroute them.

4.3 Step-by-step Analysis

Working

Given Destination IP: 192.168.2.45

Given Routing Table Entries:

- 192.168.2.0/24
- 192.168.1.0/24
- 192.168.3.0/24

1. Determine the Destination Subnet:

To determine the destination subnet, we compare entry:

- Is within ? No. 192.168.2.45 192.168.2.45
  192.168.1.0/24 against each routing table

- Is 192.168.2.45 within 192.168.2.0/24 ? Yes. The network address for is
  192.168.2.0/24
  192.168.2.0 , and the broadcast is 192.168.2.255 . falls within this range.
  192.168.2.45
- Is within ? No.
  192.168.2.45 192.168.3.0/24

Therefore, the destination subnet is .
192.168.2.0/24

2. Identify the Matching Routing Table Entry:

Based on the above, the matching routing table entry is
.
192.168.2.0/24

3. Determine the Next-Hop Network/Interface:

Since the packet's destination IP belongs to the 192.168.2.0/24 subnet, the
192.168.2.45
router will forward the packet out the interface directly connected to the 192.168.2.0/24 network. The next-hop is the interface associated with the 192.168.2.0/24 [redacted]etwork.

4. Default Route:

If the destination IP address does not match any specific entry in the routing table, the router uses a defaultroute. The standard representation for a defaultroute is .
0.0.0.0/0

Final Answer

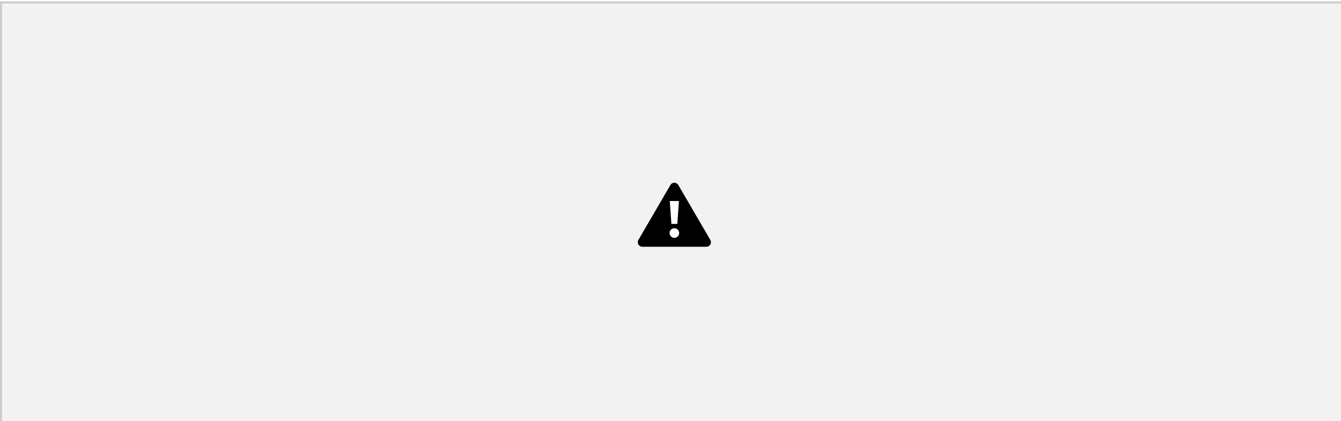
- **Destination Subnet:** 192.168.2.0/24
- **Matching Routing Table Entry:** 192.168.2.0/24
- **Next-Hop Network/Interface:** The interface directly connected to the network.

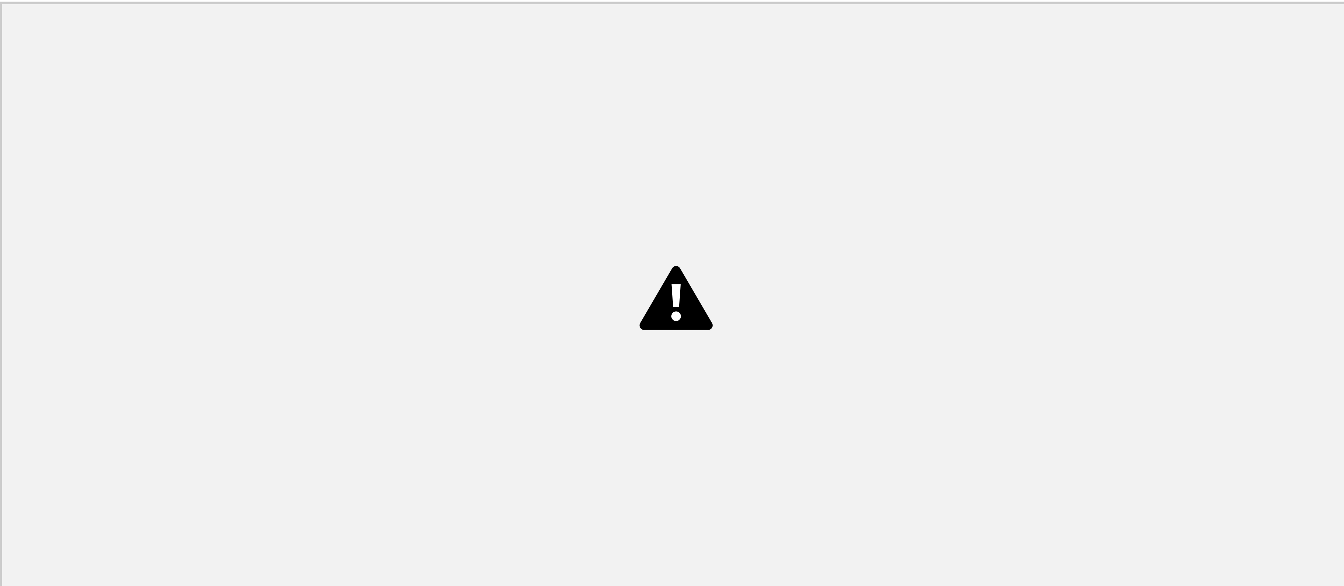
192.168.2.0/24

- **Default Route:** 0.0.0.0/0

4.4 Packet Forwarding Diagram

Figure 4.1: Router's Packet Forwarding Logic





4.5 Verification

The packet forwarding logic follows standard routing principles, specifically the longest prefix match. The destination IP clearly falls within the subnet.

192.168.2.45 192.168.2.0/24

The concept ofa defaultroute for unmatched destinations is also a fundamental routing mechanism. The Python script confirmed the matching subnet and the defaultroute.

4.6 Conclusion

This section successfully analyzed the packet forwarding process within an ISP router, demonstrating how a router determines the correct destination subnet and matching routing table entry for a given packet. The identification ofthe next-hop interface and the role of the default route are critical aspects of network routing.

5. Question 5: Layer-3 Router and LAN Connectivity

5.1 Scenario Explanation

A company establishes two office departments, Administration and Finance, connected through a Layer-3 router. The allocated subnet information is: Administration LAN ([redacted]) and Finance LAN (192.168.10.64/26). The router interfaces are configured

192.168.10.0/26

with Administration Gateway 192.168.10.1 and Finance Gateway . The task

192.168.10.65

involves determining network and broadcast addresses, valid host IP address ranges, suggesting suitable IP addresses for three computers in each LAN, and identifying the default gateway for hosts in each LAN.

5.2 Network Design Approach

This scenario focuses on the configuration and operation of a Layer-3 router acting as a gateway between two distinct LANs. Each LAN is assigned a subnet, providing 62 usable host addresses. The router's interfaces serve as the default gateways fortheir respective LANs, enabling inter-LAN communication and access to external networks. The design ensures properIP allocation, host addressing, and gateway configuration for seamless network operation.

5.3 Step-by-step Calculations

Formula Used

- **Number of Addresses in a Subnet:**
- **Number of Usable Hosts:**

Working

Given Subnets:

- Administration LAN: 192.168.10.0/26

- Finance LAN:

192.168.10.64/26 192.168.10.1

Given Router Gateways: • Administration Gateway:

• Finance Gateway:

192.168.10.65

1. Administration LAN - /26

- **Subnet Mask:** (255.255.255.192) • **Number of Addresses:**

192.168.10.0

- **Network Address**

- **Broadcast Address:** 192.168.10.63 (192.168.10.0 + 64 - 1)

- **Valid Host Range:** • **Usable Hosts:** • **Default Gateway:** 192.168.10.62 to

192.168.10.1 192.168.10.1

- **Suggested IPs for 3 computers (excluding gateway):** , ,

192.168.10.4 192.168.10.2 192.168.10.3

2. Finance LAN - /26

- **Subnet Mask:** (255.255.255.192) • **NumberofAddresses:**

192.168.10.64

- **Network Address**

- **Broadcast Address:** • **Valid Host Range:** 192.168.10.127 to (192.168.10.64 + 64 - 1)

Default Gateway:

- **Usable Hosts:** •

192.168.10.65

192.168.10.65

192.168.10.126

- **Suggested IPs for 3 computers (excluding gateway):** , ,

192.168.10.68 192.168.10.66 192.168.10.67

Final Answer

Administration LAN: •

Network Address

192.168.10.0

Address:

192.168.10.63

- **Broadcast**
  - **Valid Host IP Address Range:** 192.168.10.1 192.168.10.62

to

- **Suggested IP Addresses for 3 Computers:** 192.168.10.2 192.168.10.4 192.168.10.3

, ,

- **Default Gateway:**

Address

192.168.10.64

Finance LAN:

192.168.10.1

- **Broadcast Address:**

192.168.10.127

- **Network**
  - **Valid Host IP Address Range:** 192.168.10.126

192.168.10.65

to

- **Suggested IP Addresses for 3**

Computers:

, ,

- **Default Gateway:**

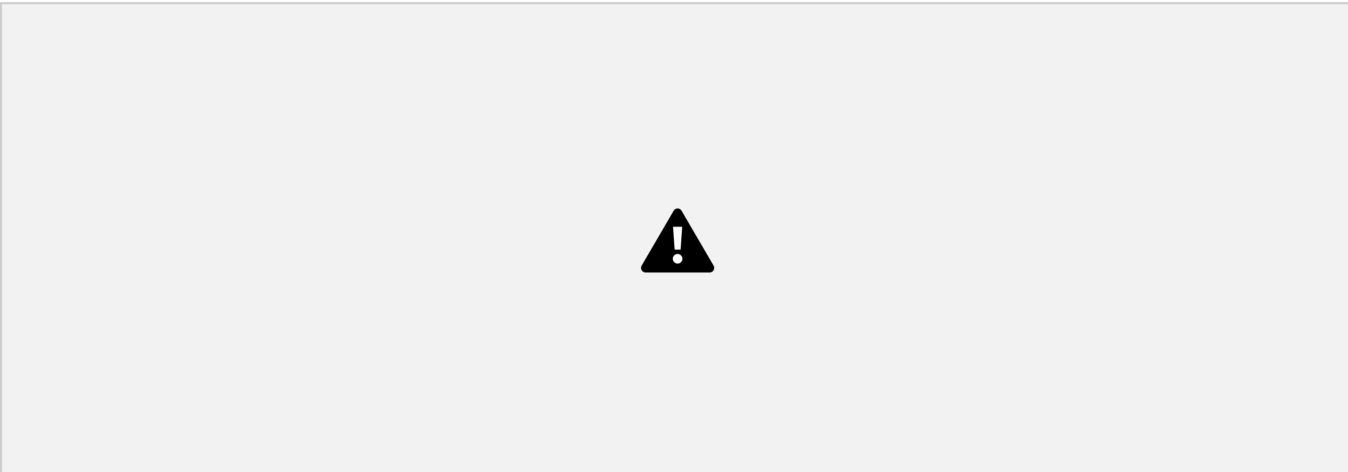
192.168.10.67

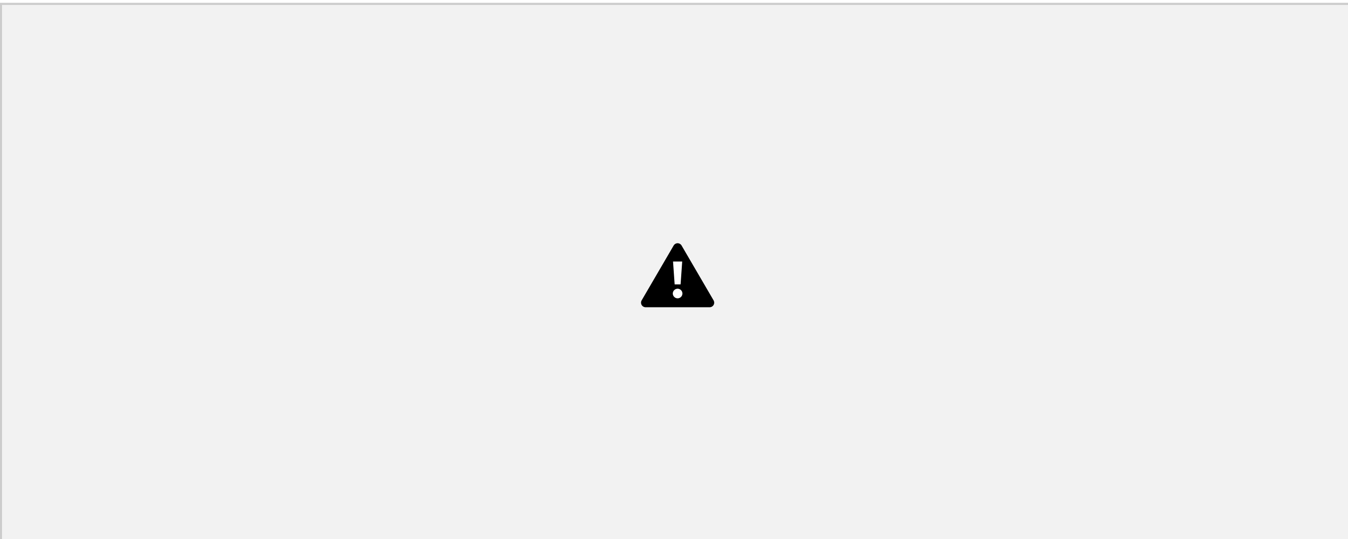
192.168.10.66 192.168.10.68

192.168.10.65

5.4 Router Diagram

Figure 5.1: Layer-3 Router Connections





5.5 IP Address Table

LAN

Administrati on

Network Address	Fir

192.168.10.0	19
192.168.10.6 4	19

CIDR /26

Finance /26 **5.6 Subnet Table**

Subnet ID 1

Network Address	C	
192.168.10.0	/2	Usable Hosts
192.168.10.64	/2	62

262 **5.7 Verification**

The subnet calculations for both Administration and Finance LANs were verified, confirming the network, broadcast, and valid hostranges. The suggested IP addresses forthe computers are within the valid host range and do not conflict with the network, broadcast, or gateway addresses. The Python script also confirmed these values.

5.8 Conclusion

This section successfully configured the IP addressing for two LANs connected via a Layer-3 router, demonstrating proper subnetting, host IP assignment, and default gateway configuration. This setup ensures that devices within each LAN can communicate with each other and with devices in other LANs via the router, forming a robust and functional network segment.

References

[1] Cisco. (n.d.). IPv6 Addressing: The Basics. Retrieved from

[2] Techopedia. (n.d.). What is Subnetting? Retrieved from

[3] GeeksforGeeks. (n.d.). VLSM (Variable Length Subnet Mask). Retrieved from [4] NetworkLessons.com. (n.d.). How Routers Forward Packets. Retrieved from